Pharmaceutical and Biopharmaceutical Processing

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Pharmaceutical and Biopharmaceutical Processing (A26741)

Short Title: Pharm & Biopharm Proc
Faculty Science and Computing
Department Science
Cluster Pharmaceutical Science
Author AHAYDEN
Credits 5 **Level:** Introductory

Description of Module / Aims

This module describes the overall operation, unit processes, and key parameters involved in the scale-up and production of pharmaceutical drug substances and biopharmaceutical products. The module includes an in-depth review of API reactors, drug crystallisation, filtration processes, drying issues and drug powder properties. The second half of the theory element of the module will focus on the production of biopharmaceutical products on an industrial scale. This will encompass the upstream and downstream activities involved in the processing of biopharmaceutical products and in particular lyophilisation and fill finish. One important aspect of the module is an examination of the importance of process control and the reduction of hazards and deviations in the industry. The course is designed to be suitable for both pharmaceutical and biopharmaceutical students.

Programmes

SCIE-0078	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	2 / 4 / M

Indicative Content

- Pharmaceutical operations: facility design, layout, process flow, process control, PAT, process safety; green chemistry
- Pharmaceutical unit operations and equipment: reactors, reactor kinetics, crystallisation, filtration, drying, powder properties
- Biopharmaceutical unit operations and equipment: fermentation, bioreactor design, scale-up, harvesting, filtration, purification
- Biopharmaceutical unit operations and equipment: filling, lyophilisation, packaging
- Pharmaceutical/biopharmaceutical facilities, documentation: BMRs, BPRs, Out of Specification results, labelling
- Aspects in scale-up and pilot plant production

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate the importance of design, layout and process control to pharmaceutical operations.
2. Illustrate and explain the operation and function of pharmaceutical and biopharmaceutical processing equipment.
3. Compare individual processes and their underlying theory and describe associated documentation.
4. Discuss the formulation and freeze-drying activities for a Biopharmaceutical process.
5. Discuss and categorise milestones and hazards in pharmaceutical scale-up.
6. Demonstrate knowledge of calculations involved in reactor kinetics.

Learning and Teaching Methods

- Lectures.

- Field trips.
- Group work.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	50%	1,2,3,4,5,6
In-Class Assessment	15%	1,6
Presentation	15%	1,2,3,4,5
Final Written Examination	50%	1,2,3,4,5

Assessment Criteria

<40%: The student will have failed to attain the learning outcomes at even a basic level.
40%-49%: The student will have attained the learning outcomes at a basic level.
50%-59%: The student will be able to demonstrate understanding of some of the complexities of the topics.
60%-69%: The student will demonstrate a more detailed knowledge of all of the topics covered, and will have the ability to interpret data.
70%-100%: The student will demonstrate a comprehensive understanding of all of the material covered, including calculations.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Independent Learning	99	

Essential Material

- Mollah, A.H., M. Long and M. Baseman. _Risk Management Applications in Pharmaceutical and Biopharmaceutical Applications_. USA: Wiley, 2013.
- Nuslim, S. _Active Pharmaceutical Ingredient. Development, Manufacture and Regulation_. USA: Taylor and Francis, 2005.

Supplementary Material

- Cox, S. _Pharmaceutical Manufacturing Handbook: Production and Processes_. USA: Wiley, 2008.
- Doble, M. and A.K. Krutheventi. _Green Chemistry and Processes_. London: Academic Press, 2007.
- Franks, F. _Freeze Drying of Pharmaceuticals and Biopharmaceuticals. Principle and Practice_. 1. London: RSC, 2007.